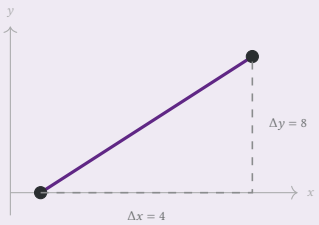
QOIDALAR VARAG'I – 1/2

QOIDA · Burchak koeffitsienti (slope)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Atama ko'prigi: an'anaviy darslikda bu k (va $y = kx + b$), SAT'da esa **slope** va m (va $y = mx + b$) deyiladi – bir xil narsa.

Misol: $(3, 2)$ va $(7, 10)$ orqali:
$m = \frac{10 - 2}{7 - 3} = \frac{8}{4} = 2$



$y = mx + b$ **da:** m – qiyalik, b – kesilgan nuqta. Ularni chalkashtirmang: $y = 5x - 3$ da qiyalik 5, kesilgan nuqta esa -3.

DIQQAT · Formulani teskari qo'llamang

Har doim **tepaga (y) o'zgarish – pastga (x) o'zgarish** tartibida: $m = \frac{y_2 - y_1}{x_2 - x_1}$, **hech qachon** $\frac{x_2 - x_1}{y_2 - y_1}$ emas.

QOIDA · To'g'ri chiziq tenglamasini yozish

Nuqta-qiyalik shakli: $y - y_1 = m(x - x_1)$

Qiyalik 4, **nuqta** $(1, 3)$:
$y - 3 = 4(x - 1) \Rightarrow y - 3 = 4x - 4 \Rightarrow y = 4x - 1$

Ikki nuqta $(2, 1)$ **va** $(4, 7)$: avval m topiladi: $m = \frac{7 - 1}{4 - 2} = 3$; so'ng $(2, 1)$ bilan:
$y - 1 = 3(x - 2) \Rightarrow y - 1 = 3x - 6 \Rightarrow y = 3x - 5$

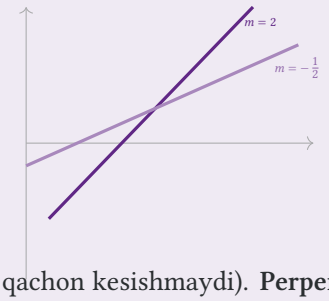
Slope-intercept shakl ($y = mx + b$) – grafik chizish va qiyalikni o'qish uchun qulay; nuqta-qiyalik shakli – tenglama yozishni boshlash uchun qulay.

QOIDALAR VARAG'I – 2/2

QOIDA · Parallel va perpendikulyar chiziqlar

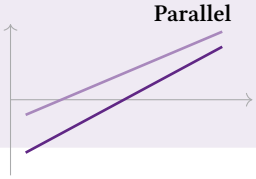
$m_1 = m_2$ (parallel) $\qquad$ $m_1 \times m_2 = -1$ (perpendikulyar)

$y = 2x - 5$ ga perpendikulyar, $(4, 1)$ dan o'tuvchi:
Qiyalik 2 ning manfiy teskarisi: $-\frac{1}{2}$.
$y - 1 = -\frac{1}{2}(x - 4) \Rightarrow y = -\frac{1}{2}x + 3$

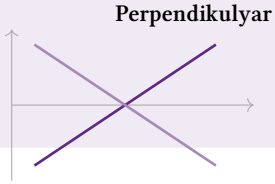


Parallel chiziqlar – bir xil qiyalik, boshqa kesilgan nuqta (hech qachon kesishmaydi). **Perpendikulyar** – to'g'ri burchak ostida kesishadi.

Parallel



Perpendikulyar



DIQQAT · Manfiy teskarini to'liq bajaring

Perpendikulyar qiyalik ham teskari, ham ishorasi qarama-qarshi bo'ladi: $\frac{2}{3} \to -\frac{3}{2}$ – faqat teskarisini olib, ishorani unutmang.

QOIDA · Qiyalik va kesilgan nuqtaning amaliy ma'nosi

Amaliy masalada: **qiyalik** = o'zgarish tezligi (har birlik uchun), **kesilgan nuqta** = boshlang'ich ($x = 0$ dagi) qiymat.

Misol: $C(x) = 2000x + 5000$ (taksi, x=km). $C(10) = 2000(10) + 5000 = 25000$ so'm. (2000 – 1 km narxi; 5000 – boshlang'ich to'lov.)

DIQQAT: savolni oxirigacha o'qing – qiyalikning o'zi so'ralyaptimi, ma'lum x dagi qiymatmi, yoki x ning o'zi so'ralyaptimi? Uchalasi ham boshqa-boshqa amal.

Ishora ham ma'noga ega: manfiy qiyalik – kamayish (masalan, suv bakdan chiqib ketayotgan suv), musbat qiyalik – o'sish.